

Final Preparation & Review

GOVT 10: Quantitative Political Analysis

This chapter reviews the concepts the exam draws on, works through three readings of printed regression output of the kind the exam uses, and lists the mistakes that cost the most points.

1 Concept Review

1.1 Design

Before fitting any model, ask what determined which units received the treatment.

Design Concepts to Have at Your Fingertips

- **Experimental vs. observational.** Experiments randomize treatment; observational studies do not.
- **Three requirements for causation.** Association, temporal ordering, no confounding.
- **Natural experiments.** Treatment is as-if random: a lottery, a redistricting line, a policy cutoff.
- **Difference-in-differences.** Compare the change in the treated group to the change in the control group. Requires parallel trends: the groups may start at different levels, but must have been moving the same way.
- **Threats to inference.** Selection bias, reverse causation, omitted variable bias, non-compliance, attrition, spillover, external validity.

The exam tests two threats that are easy to confuse. **Omitted variable bias** arises when a confounder you failed to measure drives both treatment and outcome. **Post-treatment bias** arises when you control for something the treatment itself changed. The second is the easier mistake to make, because adding controls looks like care. If a variable was measured after treatment, controlling for it removes part of the effect you are trying to estimate.

1.2 Description

Description Concepts

- **Central tendency.** Mean, median, mode. The median resists outliers; the mean does not.
- **Spread.** Range, variance, standard deviation. Standard deviation measures variability *in the data*; standard error measures variability *in an estimate*.
- **Tidyverse verbs.** `filter()` keeps rows, `select()` keeps columns, `arrange()` sorts, `mutate()` creates, `group_by()` + `summarise()` aggregates.
- **Logical conditions.** Commas inside `filter()` act like AND, not OR.
- **Missing values.** `mean(x, na.rm = TRUE)` drops the missing rows and averages what remains; it does not impute anything.
- **Plotting.** Bar charts for counts, scatter plots for two continuous variables, box plots for group comparisons, line plots for time series.

The distinction the exam tests most often is `mutate()` versus `summarise()` after a `group_by()`. Both compute the same group statistic. `summarise()` collapses to one row per group and keeps only the columns you name; `mutate()` keeps every row and every column and appends one more, so each row carries its own group's value. When a question says "keep all the original variables," it is asking for `mutate()`.

1.3 Regression

Regression Concepts

- **OLS.** The line of best fit minimizes the sum of squared residuals.
- **Coefficients.** A slope coefficient is the predicted change in Y for a one-unit increase in X, holding the other predictors constant.
- **Intercept.** The predicted Y when every predictor equals zero. Often meaningless on its own.
- **Dummies.** A categorical variable enters as one dummy for every level except the reference. Every dummy coefficient is a comparison to the omitted reference category, never to another dummy.
- **Interactions.** The interaction coefficient is the *difference* between one variable's effect at different levels of the other. The main effect is the slope when the interacting variable is zero.
- **Controls.** A control isolates a partial effect. It does not by itself establish causation, and controlling for a post-treatment variable makes the estimate worse.

- **Fit.** R^2 is the share of variance explained; RMSE is the typical size of a prediction error, in the outcome's units.

1.4 Inference

The Four Languages of a Single Result

A coefficient and its standard error can be expressed four ways:

1. Standard error: the precision of the estimate.
2. t-statistic: how many standard errors the estimate sits from zero.
3. p-value: the probability of a result at least this extreme if the true effect were zero.
4. 95% CI: the range of values the data cannot rule out.

All four come from the same two numbers, so they can never disagree. The CI is the most informative single summary, because it reports effect size, precision, and significance at once.

This equivalence is exact and worth memorizing in one line: $p < 0.05$ **if and only if the 95% confidence interval excludes zero**. If a table reports a significant p-value alongside an interval that straddles zero, one of the numbers is wrong.

2 Three Worked Reads

The exam prints regression output and asks you to read it. These three cases cover most of what it asks.

2.1 Reading an Interaction

Does the return to education differ for union members? Fit the model and read the coefficients.

```
model_int <- lm(turnout ~ education * union_member, data = union_demo)
tidy(model_int, conf.int = TRUE)
```

```
# A tibble: 4 x 7
  term                estimate std.error statistic  p.value conf.low conf.high
<chr>                <dbl>    <dbl>    <dbl>    <dbl>    <dbl>    <dbl>
1 (Intercept)         35.0      2.61     13.4  4.23e-34  29.9     40.2
2 education            2.07     0.183     11.3  5.57e-26   1.71     2.43
3 union_member        -2.68     4.18     -0.640  5.23e- 1 -10.9     5.55
4 education:union_member  0.953    0.293     3.25  1.25e- 3   0.377    1.53
```

Read it in three steps. The `education` coefficient is the slope *for non-members*, because that is what the model gives you when `union_member` is zero. The `education:union_member` coefficient is how much steeper the slope is for members. Adding the two gives the members' slope. The `union_member` coefficient is the gap between members and non-members *among people*

with zero years of education. It is not an average gap, and on its own it is rarely a quantity of interest.

2.2 Reading a Difference-in-Differences Table

Four states adopted a registration reform; four did not. The 2x2 table of group means is all you need.

```
did_table <- tibble(  
  group = c("Adopted", "Adopted", "Did not adopt", "Did not adopt"),  
  year   = c(2016, 2022, 2016, 2022),  
  regist = c(41.2, 52.8, 44.6, 49.1)  
)  
  
did_table %>%  
  pivot_wider(names_from = year, values_from = regist) %>%  
  mutate(change = `2022` - `2016`)
```

```
# A tibble: 2 x 4  
  group      `2016` `2022` change  
  <chr>      <dbl> <dbl> <dbl>  
1 Adopted      41.2   52.8   11.6  
2 Did not adopt 44.6   49.1    4.5
```

The treated group gained 11.6 points and the control group gained 4.5. The difference-in-differences estimate is $11.6 - 4.5 = 7.1$ percentage points. The control group's 4.5 estimates everything else that happened over those six years, and the design works by subtracting it.

Note that the adopting states started 3.4 points *lower*. Difference-in-differences does not require the groups to match in level, only that, absent the policy, they would have moved in parallel.

2.3 Reading a Confidence Interval Off Regression Output

```
model_ci <- lm(approval ~ news_hours + age, data = ci_demo)  
confint(model_ci)
```

```
                2.5 %      97.5 %  
(Intercept) 38.2519678 49.45643637  
news_hours   0.6064139  1.78076893  
age          -0.1867064 -0.01663216
```

Compare each interval to zero. An interval that excludes zero corresponds to $p < 0.05$; one that contains zero does not. But width carries information the p-value discards: a narrow interval sitting just across zero is a precisely estimated near-null, while a wide interval spanning large negative and large positive values means the study had nothing to say. Both report “not significant,” but they are different findings.

3 Approaching the Final Exam

3.1 What the Exam Asks For

Two hours, closed book, on paper, all questions worth equal points.

Question type	Count
Multiple choice	44
TRUE/FALSE	6
Predicted value from a printed regression table	4
Difference-in-differences estimate	1
Fill in the blank	1
Matching	2
Total	58

Table 1: The final exam by question type

The only arithmetic on the exam is the four predicted values and the one difference-in-differences estimate. You will not construct a confidence interval by hand, convert a coefficient and standard error into a t-statistic, or compute a standard error. Everything else is read off the page.

Exam Strategy

1. **Ask what the question is testing.** Design, description, regression, or inference; placing it usually narrows the answer.
2. **Watch the verb.** "Causes" requires a design that licenses it. "Is associated with" is the honest verb for observational regression.
3. **Use the CI-p equivalence.** Given an interval, you know the significance. Given a p-value, you know whether the interval crosses zero.
4. **Read the variable definitions before the question.** Income in thousands of dollars is the single most expensive detail to miss.
5. **Walk interactions term by term.** Set the dummy, write out every term including the ones that vanish, then add.
6. **Sanity-check predicted percentages.** If a predicted turnout comes out at 340 or -12 , go back and find the unit error.
7. **Show your work on the calculations.** A written-out prediction with an arithmetic slip still earns most of the points; partial credit exists only where work is shown.

3.2 Common Pitfalls to Avoid on the Final

The Top Ten Pitfalls

The first three cost the most points and are explained in full; the rest are listed briefly.

1. **Calling an observational coefficient causal without a design that supports it.** A significant slope in survey data is an *association*. Upgrading it to a cause requires random assignment or a credible natural experiment.
2. **Saying there is a 95% probability the true value lies in the interval.** The parameter is fixed; the interval is what varies. The 95% describes the long-run reliability of the procedure, not the probability that this particular interval caught the truth.
3. **Confusing standard deviation with standard error.** SD describes how much individuals differ from one another. SE describes how much an estimate would bounce across repeated samples. The standard error shrinks as the sample grows, which is why a sample can have huge spread and a precisely estimated mean at the same time.
4. Treating "not significant" as "no effect."
5. Eyeballing overlapping confidence intervals to declare a difference non-significant.
6. Reporting a statistically significant but substantively trivial effect from a very large sample as important.
7. Controlling for a post-treatment variable and erasing the mechanism.
8. Interpreting a dummy coefficient as a comparison to a group other than the reference category.
9. Reporting an interaction coefficient as one group's effect rather than the difference between two effects.
10. Running many tests and reporting the one that came out significant.

4 A Note on Causation

The habit worth carrying out of this course is a pause before the word "cause." A regression coefficient, however significant, is consistent with many causal explanations, and choosing among them requires knowing how treatment was assigned.

The tools for credible causal claims are randomized experiments (Week 3), natural experiments and difference-in-differences (Weeks 4–5), control variables in regression (Weeks 5 and 7), and an inferential vocabulary for stating what you know and what you do not (Weeks 8–9). When the design supports a causal reading, take it. When it does not, write "associated with" and say what evidence would upgrade the claim.

A Rule of Thumb

Before writing "X causes Y," ask what you would have to believe about the assignment of treatment for that to be true. If the answer requires heroic assumptions, soften the verb. If the answer is "treatment was randomly assigned and compliance was high," you are on solid ground.

5 Using AI for This Material

Where AI Helps This Week

- Generating additional practice problems in a format you specify: "give me five regression tables with an interaction and a binary moderator, and ask me for a predicted value from each."
- Checking your arithmetic on a predicted value after you have written out the terms yourself.
- Turning a coefficient and its interval into a one-sentence result you could put in a memo, so you can compare it to your own phrasing.

Where AI Gets This Wrong This Week

- Asked whether an interval containing zero means "no effect," models routinely answer yes. The correct answer is that the data are consistent with a range of effects that happens to include zero.
- Asked whether two overlapping confidence intervals imply no significant difference, models usually say yes. Overlapping intervals frequently accompany a significant difference; the comparison requires a test on the difference itself.
- Asked to "improve" a model by adding controls, models will add post-treatment variables without objection. They optimize for fit, and when a control was measured is not visible in the code.

A prompt that works for this week:

Write five multiple-choice questions that give me a regression table with a dummy variable and ask what the dummy coefficient compares to. Do not give me the answers until I have committed to mine.

A prompt that misleads:

My 95% confidence interval is $[-0.5, 3.2]$. Does this mean my treatment had no effect?

Why it misleads: the question invites a yes-or-no verdict, and models supply one. The interval says the data cannot rule out zero, but it also cannot rule out an effect of 3, which may be substantively large. Flattening that into “no effect” discards what the interval reports.

5.1 R Functions Reference

Task	Code
Filter rows	<code>filter(df, age > 18, !is.na(vote))</code>
Keep columns	<code>select(df, age, vote, party)</code>
Create a variable	<code>mutate(df, log_income = log(income))</code>
Recode categories	<code>case_when(age < 30 ~ "young", TRUE ~ "older")</code>
Group and collapse	<code>df %>% group_by(state) %>% summarise(m = mean(x))</code>
Group and keep all rows	<code>df %>% group_by(state) %>% mutate(m = mean(x))</code>
Linear regression	<code>lm(y ~ x1 + x2, data = df)</code>
Interaction model	<code>lm(y ~ x1 * x2, data = df)</code>
Tidy coefficients with CIs	<code>tidy(model, conf.int = TRUE)</code>
Whole-model fit	<code>glance(model)</code>
Regression CIs	<code>confint(model)</code>
One-sample t-test	<code>t.test(x, mu = value)</code>
Two-sample t-test	<code>t.test(y ~ group, data = df)</code>

6 Practice Problems

Reading Output

1. A 95% confidence interval for a regression coefficient is $[0.5, 3.2]$. What do you know about the p-value, about significance at $\alpha = 0.05$, and about how a 99% interval on the same data would compare in width?
2. Two studies estimate the same effect. Study A ($n = 50$) reports a coefficient of 5.0 with 95% CI $[0.8, 9.2]$. Study B ($n = 500$) reports a coefficient of 2.0 with 95% CI $[1.4, 2.6]$. Which provides stronger evidence that an effect exists? Which suggests a larger effect? What could explain the gap?
3. A regression reports a coefficient with $p = 0.008$ and a 95% CI of $[-0.2, 1.9]$. Explain why these two numbers cannot both be right.
4. In a model with region entered as dummies and the Northeast omitted, the coefficient on `region_west` is -4.5 and the coefficient on `region_south` is $+6.2$. A student writes: “Western respondents score 4.5 points below Southern respondents.” Correct the sentence.

Design and Interpretation

5. A randomized experiment finds that mailing voters a turnout reminder raised turnout by 3 percentage points (95% CI: 0.5 to 5.5, $p = 0.02$, $n = 1,000$). Write a four-sentence summary covering the causal claim, the effect size, the uncertainty, and what the effect means in practice.
6. A researcher estimates the effect of a randomly assigned mentoring program on college graduation. She then adds "credits completed in the first year" as a control, and the program coefficient falls to near zero. What has she done, and which specification should she report?
7. A study regresses test scores on study hours and class size and finds a coefficient of 3.2 on study hours (95% CI: $[2.1, 4.3]$) and -0.4 on class size (95% CI: $[-0.9, 0.1]$). Interpret each coefficient, state its significance, and say what can and cannot be concluded about causation.
8. Cities that expanded bus service saw ridership rise 14% while cities that did not saw it rise 5%. What is the difference-in-differences estimate, what assumption makes it causal, and name one thing that would break that assumption.

For Further Study

The methods here generalize. Robust standard errors handle violations of the regression assumptions. Bootstrapping builds confidence intervals without distributional assumptions. Bayesian credible intervals offer a framework in which probability statements about parameters are meaningful. Power analysis runs the same formulas in reverse to choose a sample size before collecting data. Instrumental variables, regression discontinuity, and matching extend causal inference to settings without random assignment. Each builds on the same foundations covered here: design, description, regression, and inference.